2026

(NEP—2020)

(6th Semester)

PSYCHOLOGY (MAJOR)

(Statistics in Psychology)

Full Marks : 75

Time : 3 hours

The figures in the margin indicate full marks for the questions.

Simple Calculator may be used in this paper.

(SECTION : A—OBJECTIVE)

(*Marks* : 10)

Tick (✓) the correct answer in the brackets provided : 1×10=10

1. Statistical inferences are usually made from _____ to populations.

(a) samples ( )

(b) census ( )

(c) parameters ( )

(d) variables ( )

2. A _____ is a characteristic that may take on different values.

(a) constant ( )

(b) frequency ( )

(c) parameter ( )

(d) variable ( )

3. A platykurtic distribution is characterized by

(a) a sharper peak than normal ( )

(b) heavy tails ( )

(c) a flatter peak than normal ( )

(d) perfect symmetry ( )

4. The arithmetic mean of 2, 4, 6 and 8 is

(a) 4 ( ) *(b)* 5 ( )

(c) 6 ( ) *(d)* 7 ( )

5. A parametric statistical test which measures the linear relationship between two variables is called

(a) chi-square test ( )

(b) *t*-test ( )

(c) Pearson's correlation coefficient ( )

(d) ANOVA ( )

6. A/An _____ is stated by how many standard deviations the corresponding raw score lies above or below the mean of the distribution.

(a) percentile rank ( )

(b) *T*-score ( )

(c) *Z*-score ( )

(d) individual score ( )

7. A percentile rank transformation provides information about

(a) the actual value of the score ( )

(b) the relative position of a score in the distribution ( )

(c) standard deviation only ( )

(d) mean of the scores ( )

8. The concept of small sample size test was developed by

(a) David Wechsler ( )

(b) Charles Spearman ( )

(c) William Sealy Gosset ( )

(d) Francis Galton ( )

9. The value of a correlation coefficient (r) ranges from

(a) 0 to 1 ( )

(b) -1 to $+1$ ( )

(c) $-\infty$ to $+\infty$ ( )

(d) 0 to 100 ( )

10. A _____ is a prediction intended to be tested in a research study.

(a) tip-of-the-tongue phenomenon ( )

(b) hypothesis ( )

(c) priming ( )

(d) sensory register ( )

(SECTION : B—SHORT ANSWERS)

(*Marks* : 25)

Write short notes on *five* of the following, taking at least *one* from each Unit : 5×5=25

UNIT—I

1. Descriptive and inferential statistics

2. Skewness and kurtosis

3. Measures of central tendency

UNIT—II

4. Chi-square

5. Mann-Whitney *U*-test

6. SPSS

UNIT—III

7. Types of correlation

8. Assumptions of *t*-test

9. Types of error

(SECTION : C—DESCRIPTIVE)

(*Marks* : 40)

Answer *four* of the following questions, taking at least *one* from each Unit : 10×4=40

UNIT—I

1. Explain the different scales of measurement using suitable examples.

2. Write about the properties and applications of normal probability curve (NPC).

3. Calculate the standard deviation of the following set of scores :

Class Interval	*Frequency*
38–40	3
36–38	4
34–36	9
32–34	10
30–32	8
28–30	6
26–28	7
24–26	5

UNIT—II

4. Compare and contrast between parametric and non-parametric statistics.

5. A researcher studied the daily practice hours of two independent groups of students preparing for an exam. Their scores were as follows :

Less than 2 hours	4	6	2	5	8	3	1	?
More than 2 hours	9	11	7	6	14	13	12	10

Apply the Mann-Whitney *U*-test to examine whether there is a significant difference in practice hours between the two groups. Test the significance at the 0·05 level. State the null hypothesis and conclusion. 6+2+2=10

6. The following data shows the concentration score of students in three different study environments :

Library	*Home*	*Café*
15	11	8
16	12	9
14	10	7
17	13	6
15	11	8
16	12	9

Compute the Kruskal-Wallis *H*-statistic test at 0·05 level of significance and interpret the results. 8+2=10

UNIT—III

7. What is hypothesis testing? Discuss one-tailed and two-tailed tests of significance. 4+6=10

8. A psychologist wants to investigate the relationship between the number of hours spent on social support (X) each day and the self-esteem (Y) of college students. The data from 10 students are shown below :

Students	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
Score on X	20	22	24	26	28	30	32	34	36	38
Score on Y	25	30	32	35	38	40	43	46	50	55

Compute Pearson's r at the 0·05 level and interpret the results. 8+2=10

9. A psychologist studied the effect of three different teaching methods on students' achievement scores. The scores obtained are given below :

Method A	65	68	70	72
Method B	60	62	64	66
Method C	70	72	74	76

Using one-way ANOVA, test whether there is a significant difference among the three teaching methods at the 0·05 level of significance. State your conclusion. 8+2=10

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